NeuroSpace

Complete evaluation export

Job ID	a67321c8-ec76-4a6d-b191-65313aca1327
Status	completed
Result timestamp	2026-06-07T00:55:49.575468
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Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	20
manual_overrides.qrnn_epochs	10
manual_overrides.batch_size	32
manual_overrides.rnn_batch_size	32
manual_overrides.qrnn_batch_size	16
num_classes	6
class_names	1, 2, 3, 4, 5, 6
qrnn_backend	qiskit
noise_enabled	True
noise_level	0.0010
mitigation_enabled	True
mitigation_runs	3

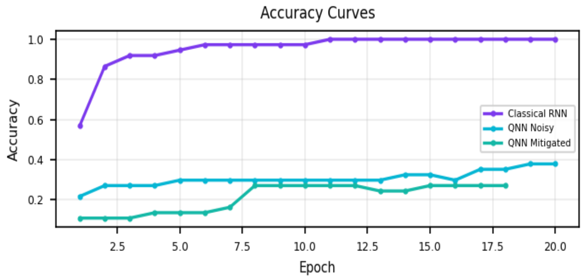
Best Model

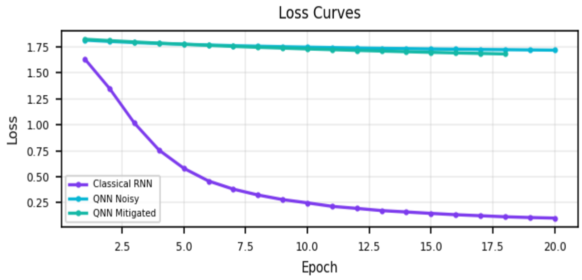
Winner	RNN Accuracy	QNN Accuracy	Improvement
RNN	94.59%	39.19%	55.41%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss	Training Time
Classical RNN	94.59%	95.44%	94.59%	94.28%	0.1005	0.58s
QNN Noisy	39.19%	43.26%	39.19%	35.97%	1.7189	54.44s
QNN Mitigated	36.49%	41.55%	36.49%	34.35%	1.6821	2m 30.96s

Learning Curves


Accuracy Curves
Accuracy
Epoch
Classical RNN
QNN Noisy
QNN Mitigated
1.0
0.8
0.6
0.4
0.2
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


Loss Curves
Loss
Epoch
Classical RNN
QNN Noisy
QNN Mitigated
1.75
1.50
1.25
1.00
0.75
0.50
0.25
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

Classical RNN

Metric	Value
Accuracy	94.59%
Precision	95.44%
Recall	94.59%
F1 Score	94.28%
Loss	0.1005
Training Time	0.58s
Train Loss	0.1209
Val Loss	0.1005
Train Accuracy	98.43%
Val Accuracy	100.00%
Train Size	255
Val Size	37
Test Size	74

Classical RNN Confusion Matrix

Actual / Predicted	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Class 1	23	0	0	0	0	0
Class 2	0	8	0	2	1	1
Class 3	0	0	15	0	0	0
Class 4	0	0	0	10	0	0
Class 5	0	0	0	0	10	0
Class 6	0	0	0	0	0	4

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
1	100.00%	100.00%	100.00%	23.0000
2	100.00%	66.67%	80.00%	12.0000
3	100.00%	100.00%	100.00%	15.0000
4	83.33%	100.00%	90.91%	10.0000
5	90.91%	100.00%	95.24%	10.0000
6	80.00%	100.00%	88.89%	4.0000
macro avg	92.37%	94.44%	92.51%	74.0000
weighted avg	95.44%	94.59%	94.28%	74.0000
Accuracy			94.59%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.7352	1.6320	30.20%	56.76%
2	1.3074	1.3480	60.00%	86.49%
3	0.9797	1.0157	80.39%	91.89%
4	0.7656	0.7552	81.18%	91.89%
5	0.6255	0.5802	90.59%	94.59%
6	0.5179	0.4579	89.80%	97.30%
7	0.4633	0.3797	93.33%	97.30%
8	0.3817	0.3231	93.33%	97.30%
9	0.3364	0.2787	93.33%	97.30%
10	0.2989	0.2474	94.51%	97.30%
11	0.2641	0.2136	97.65%	100.00%
12	0.2267	0.1939	97.25%	100.00%
13	0.2191	0.1727	96.47%	100.00%
14	0.2045	0.1595	96.86%	100.00%
15	0.1729	0.1459	97.25%	100.00%
16	0.1455	0.1332	98.43%	100.00%
17	0.1482	0.1234	97.65%	100.00%
18	0.1394	0.1132	98.43%	100.00%
19	0.1080	0.1060	98.82%	100.00%
20	0.1209	0.1005	98.43%	100.00%

QNN Noisy

Metric	Value
Accuracy	39.19%
Precision	43.26%
Recall	39.19%
F1 Score	35.97%
Loss	1.7189
Training Time	54.44s
Train Loss	1.6501
Val Loss	1.7189
Train Accuracy	43.92%
Val Accuracy	37.84%
Train Size	255
Val Size	37
Test Size	74

QNN Noisy Confusion Matrix

Actual / Predicted	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Class 1	7	0	14	2	0	0
Class 2	0	3	0	0	8	1
Class 3	3	0	11	0	1	0
Class 4	0	2	0	0	7	1
Class 5	0	0	2	0	7	1
Class 6	0	1	0	0	2	1

QNN Noisy Classification Report

Class	Precision	Recall	F1 Score	Support
1	70.00%	30.43%	42.42%	23.0000
2	50.00%	25.00%	33.33%	12.0000
3	40.74%	73.33%	52.38%	15.0000
4	0.00%	0.00%	0.00%	10.0000
5	28.00%	70.00%	40.00%	10.0000
6	25.00%	25.00%	25.00%	4.0000
macro avg	35.62%	37.29%	32.19%	74.0000
weighted avg	43.26%	39.19%	35.97%	74.0000
Accuracy			39.19%	

QNN Noisy Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.8182	1.8156	22.35%	21.62%
2	1.7961	1.8030	23.14%	27.03%
3	1.7793	1.7919	23.92%	27.03%
4	1.7645	1.7833	30.20%	27.03%
5	1.7547	1.7754	31.37%	29.73%
6	1.7426	1.7686	32.16%	29.73%
7	1.7340	1.7618	32.94%	29.73%
8	1.7229	1.7569	33.73%	29.73%
9	1.7142	1.7521	34.90%	29.73%
10	1.7000	1.7476	35.69%	29.73%
11	1.7002	1.7436	35.69%	29.73%
12	1.6904	1.7394	33.73%	29.73%
13	1.6823	1.7362	33.73%	29.73%
14	1.6781	1.7337	32.55%	32.43%
15	1.6774	1.7304	32.55%	32.43%
16	1.6723	1.7280	32.55%	29.73%
17	1.6644	1.7252	36.08%	35.14%
18	1.6566	1.7234	42.35%	35.14%
19	1.6531	1.7211	43.92%	37.84%
20	1.6501	1.7189	43.92%	37.84%

QNN Mitigated

Metric	Value
Accuracy	36.49%
Precision	41.55%
Recall	36.49%
F1 Score	34.35%
Loss	1.6821
Training Time	2m 30.96s
Train Loss	1.6289
Val Loss	1.6821
Train Accuracy	40.78%
Val Accuracy	27.03%
Train Size	255
Val Size	37
Test Size	74

QNN Mitigated Confusion Matrix

Actual / Predicted	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
Class 1	12	0	11	0	0	0
Class 2	0	3	5	0	0	4
Class 3	6	0	8	0	1	0
Class 4	0	2	6	0	0	2
Class 5	0	0	9	0	1	0
Class 6	0	1	0	0	0	3

QNN Mitigated Classification Report

Class	Precision	Recall	F1 Score	Support
1	66.67%	52.17%	58.54%	23.0000
2	50.00%	25.00%	33.33%	12.0000
3	20.51%	53.33%	29.63%	15.0000
4	0.00%	0.00%	0.00%	10.0000
5	50.00%	10.00%	16.67%	10.0000
6	33.33%	75.00%	46.15%	4.0000
macro avg	36.75%	35.92%	30.72%	74.0000
weighted avg	41.55%	36.49%	34.35%	74.0000
Accuracy			36.49%	

QNN Mitigated Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.8218	1.8236	19.22%	10.81%
2	1.8019	1.8102	19.61%	10.81%
3	1.7854	1.7978	19.22%	10.81%
4	1.7688	1.7860	20.78%	13.51%
5	1.7572	1.7757	20.78%	13.51%
6	1.7421	1.7652	21.18%	13.51%
7	1.7293	1.7550	25.88%	16.22%
8	1.7119	1.7463	29.80%	27.03%
9	1.7064	1.7385	35.29%	27.03%
10	1.6928	1.7307	35.69%	27.03%
11	1.6823	1.7238	34.51%	27.03%
12	1.6738	1.7166	35.69%	27.03%
13	1.6634	1.7101	37.25%	24.32%
14	1.6603	1.7038	40.00%	24.32%
15	1.6474	1.6979	36.47%	27.03%
16	1.6427	1.6924	35.69%	27.03%
17	1.6317	1.6874	37.65%	27.03%
18	1.6289	1.6821	40.78%	27.03%